CS-010 CF Grounding & Electrical Bonding

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Title	Carbon Fiber Grounding & Electrical Bonding
Revision	C
Status	Draft, pending Lead signature
Effective date	2026-08-28
Supersedes	Tribal grounding knowledge from SN5 trials

Approvals

Role	Name	Date
Author	Simon Starbuck (drafted with Claude, SN6 Resources)	2026-07-12
Reviewer	simon (reviewer agent)	
Approver (Composites Lead)		

Revision history

Rev	Date	Author	Description of change
A	2026-07-12	Claude	Initial issue, encodes the diffuser-validated method + the catch-can decision lesson
B	2026-07-28	Claude	Language pass: fewer em dashes for rhythm variety, no content change.
C	2026-08-28	Claude	Engineering pass (CS-000 Rev C conventions). Figure 1 draws the diffuser recipe in plan: mesh sheets, the 25 mm overlap, the terminable tail, and the probe pattern that proves the part. One intensifier cut from §1. No rule changed.

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1. Purpose

EV rules want conductive parts near HV/chassis grounded (our SN5 working criterion: **<5 Ω to the grounding point**). SN5 gave us both the success story and the cautionary tale. The **diffuser** (target and method written down up front, copper mesh in the stack) validated at **<1 Ω continuous everywhere** (2026-06-21) and drew praise from the design judges at comp. The **catch can** had no written criterion; its method was improvised for six months (Nov→May) and never hit target with CF wrapping, so we finally solved it by *giving up on CF for that geometry* and going to epoxy + copper paths (PP-05). This doc exists so we repeat the first one.

2. Scope

Any CF part with a grounding/bonding requirement, and any conductive-path retrofit on existing parts. The Ω target and rule interpretation come from the EV rules owner (powertrain/rules lead); this standard covers how composites *achieves and proves* the target, not what the target legally is.

3. References

Ref	Document	Where
R1	PP-05 RCA (catch-can saga)	FEB-COMP-PP-001
R2	CS-002 (mesh in the ply table)	CS-002 §7.1
R3	SN5 grounding trials: graphite powder judged insufficient vs copper mesh (Nico, #composites, Nov 2025)	Slack record
R4	Diffuser validation <1 Ω (post-comp recap)	#composites, 2026-06-21

4. Definitions

- **Grounding point**: the fastener/tab where the part's conductive path meets chassis ground.
- **Probe pattern**: the defined set of surface points measured against the grounding point.
- **CF as conductor**: carbon laminate conducts, but through-thickness resistance through resin-rich layers is high and unpredictable; bare CF also can't be reliably terminated. Hence copper mesh.

5. Equipment & Materials

Item	Spec / product	Source	Notes
Copper mesh	Fine woven mesh (repeat team buy)	Amazon	The validated conductor (R3, R4)
Multimeter	2-wire, known-good leads	Electrical	Zero the leads first (touch together, note offset)
Ring terminals / grounding hardware	Per part design	McMaster	
Graphite powder	1–5 wt% in resin	Sigma-Aldrich	Secondary/supplement only; insufficient alone (R3)

6. Safety

Grounding verification happens on or near the car: follow EV safety rules. No probing near energized HV; HV isolation confirmed by the ESO before resistance measurements on installed parts. Bench measurements on loose parts: no special hazards beyond CS-009 dust rules if sanding contact points.

7. Procedure

7.1 Decision point first (the catch-can lesson)

1. Before any layup work, answer on the WO: **is CF-with-copper-mesh the right conductive path for this geometry?**

- **Good fit:** large open laminates (diffuser, panels) where mesh lays flat in the stack and terminates at accessible flange points.
- **Bad fit:** thin-walled, tapered, or wrapped metal substrates (the catch can: sanding punctured the can wall at the taper; through-thickness contact never stabilized). For these, use **epoxy + copper strip/braid paths bonded to the surface**, or reconsider whether the part needs CF at all.

2. Write the acceptance criterion on the WO **before work starts**: Ω target, probe pattern (grid spacing or named points), and measurement method (7.3). Six months of drift happened because none of this was written (R1).

7.2 Build the path (mesh-in-stack method: the diffuser recipe)

1. Put copper mesh in the ply table per CS-002: mesh layer(s) positioned to be continuous from part surface zones to the grounding point(s), **25 mm minimum overlap** where mesh sheets meet.
2. Mesh must reach the grounding point as bare, terminable material: leave a mesh tail at the flange/fastener location, protected during infusion (peel-ply window or masked zone).
3. For surfaces needing exposed conductivity, an 88-gsm outer ply over mesh keeps finish while staying thin; validate on the first article that surface-to-ground still passes.
4. Graphite powder (1–5 wt% in the resin) may be added as a *supplement* in zones where mesh coverage is marginal, never as the primary path (SN5 trials judged it insufficient alone, R3). If used, record wt% on the WO.
5. Infuse/laminate per CS-006/CS-007. Note on the WO where mesh actually ended up (as-built).

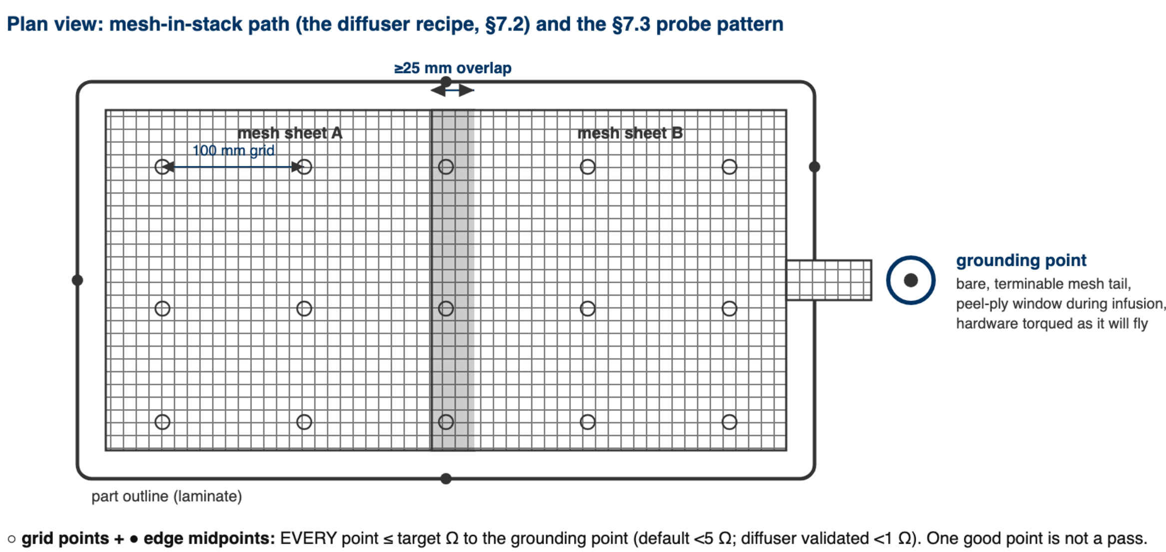

Figure 1: The mesh-in-stack path in plan: sheets overlapping 25 mm, a bare terminable tail at the grounding point, and the default probe pattern (100 mm grid plus edge midpoints). Every point measures to the grounding point; every point passes or the part does not.

[PHOTO: copper mesh tail at grounding point before bagging]

7.3 Measure: one method, every time

1. Zero the multimeter leads; record lead resistance.
2. Fasten the grounding hardware as it will fly (torqued, washers as designed); hand-held contact readings are noise.

3. Probe every point in the WO's probe pattern (default: 100 mm grid over conductive zones + every edge midpoint) to the grounding point. Record each value minus lead offset.
4. Pass = **every** point target (default <5 Ω, or the rules value on the WO). Not "one good point": that ambiguity burned a month in May 2026 (R1).

7.4 If it fails

1. Map the failing zone; usually a mesh discontinuity or resin-isolated overlap.
2. Fixes in order of preference: expose + bond a copper strip bridge (epoxy + copper path), add an external ground tab at the failing zone, or lightly abrade to mesh at a termination point and re-terminate. **Do not sand thin laminates chasing contact.** That is how the catch cans died (R1).
3. Re-measure the full pattern after any fix, not just the failed point.

8. Quality checks / acceptance criteria

Check	Target	Method	Record where
Criterion written before work	Ω + probe pattern + method on WO	WO blocker step	WO
Decision point answered	CF-mesh vs alternative path, with one-line why	WO field	WO
Resistance	Every probe point target	§7.3 method	WO qualityChecks (full table of readings)
As-built mesh record	Matches or deviations noted	CS-002 §7.4	WO

9. Records

The WO keeps: the decision-point answer, criterion, full probe-reading table (values, not "pass"), photos of mesh placement and termination, and any retrofit fixes. The diffuser's <1 Ω table is the reference example once WO-SN5 retro-records are loaded.